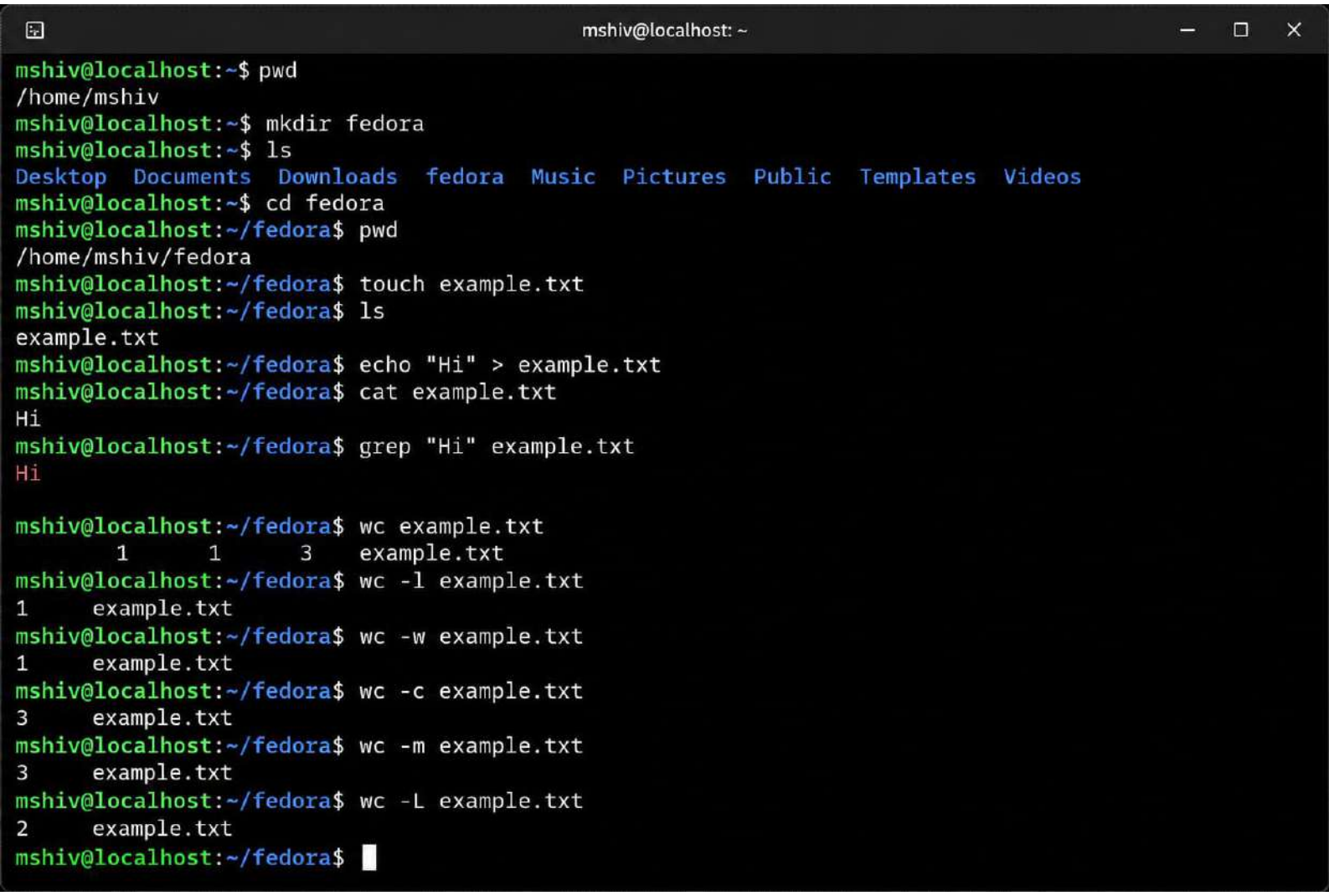
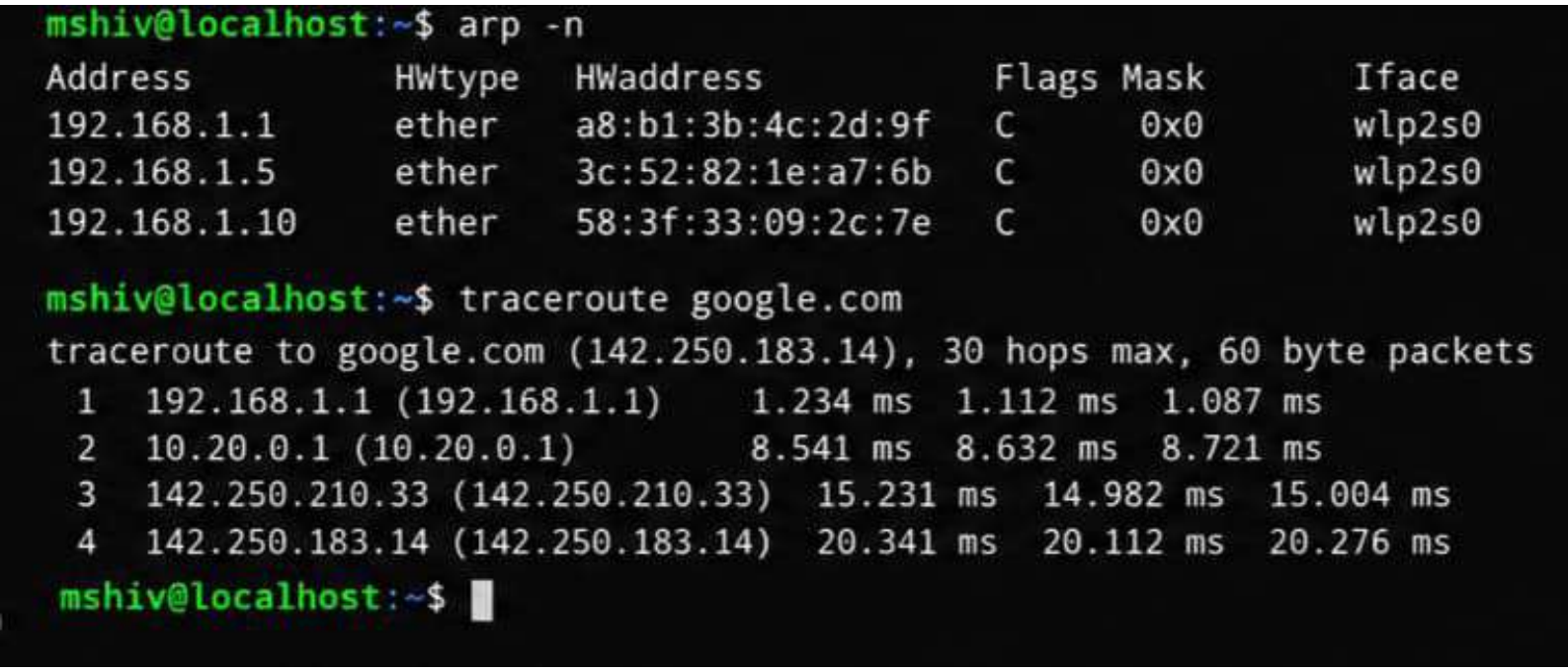
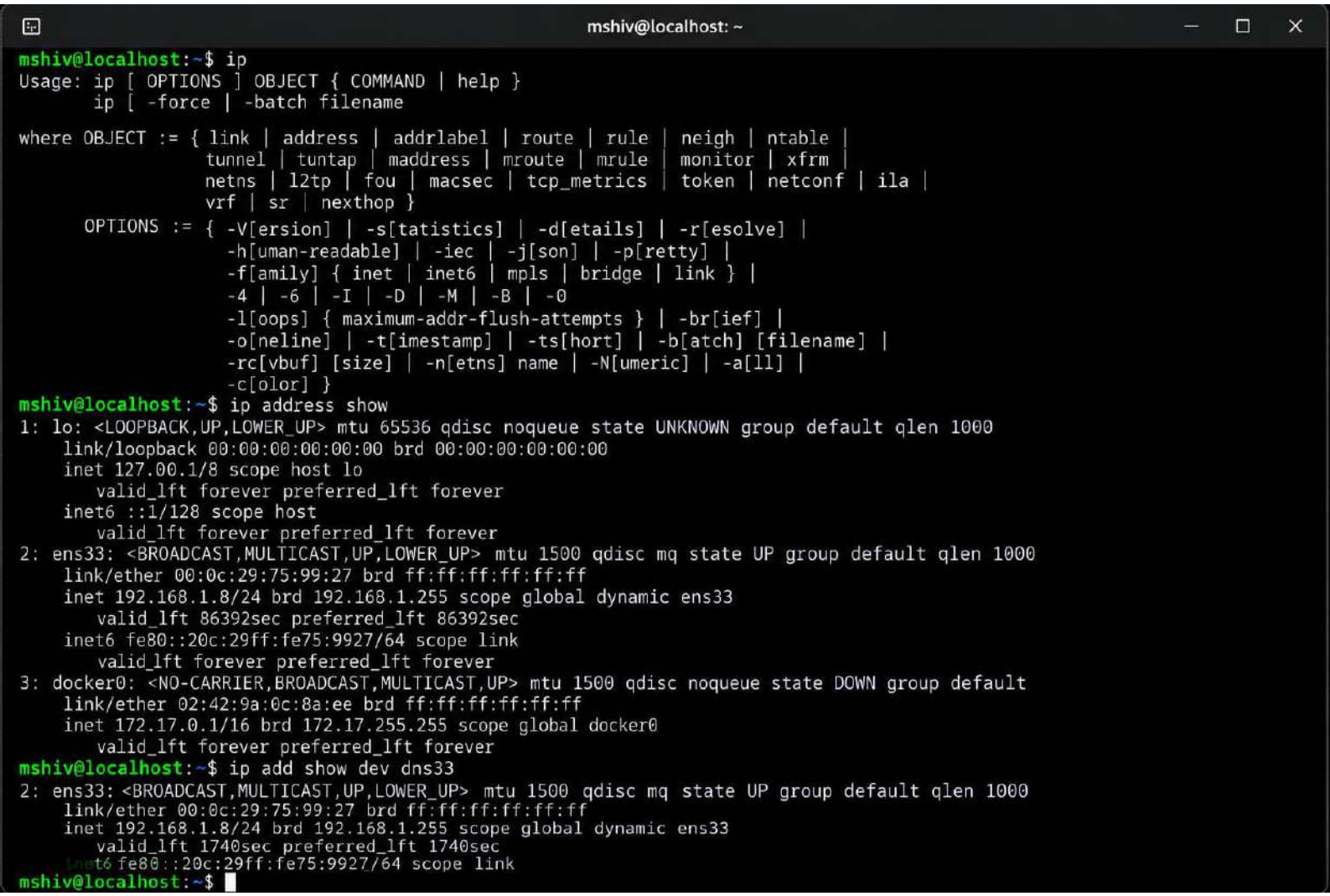

mshiv@localhost: ~
mshiv@localhost:~$ pwd
/home/mshiv
mshiv@localhost:~$ mkdir fedora
mshiv@localhost:~$ ls
Desktop Documents Downloads fedora Music Pictures Public Templates Videos
mshiv@localhost:~$ cd fedora
mshiv@localhost:~/fedora$ pwd
/home/mshiv/fedora
mshiv@localhost:~/fedora$ touch example.txt
mshiv@localhost:~/fedora$ ls
example.txt
mshiv@localhost:~/fedora$ echo "Hi" > example.txt
mshiv@localhost:~/fedora$ cat example.txt
Hi
mshiv@localhost:~/fedora$ grep "Hi" example.txt
Hi
mshiv@localhost:~/fedora$ wc example.txt
1 1 3 example.txt
mshiv@localhost:~/fedora$ wc -l example.txt
1 example.txt
mshiv@localhost:~/fedora$ wc -w example.txt
1 example.txt
mshiv@localhost:~/fedora$ wc -c example.txt
3 example.txt
mshiv@localhost:~/fedora$ wc -m example.txt
3 example.txt
mshiv@localhost:~/fedora$ wc -L example.txt
2 example.txt
mshiv@localhost:~/fedora$


mshiv@localhost:~$ arp -n
Address HWtype HWaddress Flags Mask Iface
192.168.1.1 ether a8:b1:3b:4c:2d:9f C 0x0 wlp2s0
192.168.1.5 ether 3c:52:82:1e:a7:6b C 0x0 wlp2s0
192.168.1.10 ether 58:3f:33:09:2c:7e C 0x0 wlp2s0
mshiv@localhost:~$ traceroute google.com
traceroute to google.com (142.250.183.14), 30 hops max, 60 byte packets
1 192.168.1.1 (192.168.1.1) 1.234 ms 1.112 ms 1.087 ms
2 10.20.0.1 (10.20.0.1) 8.541 ms 8.632 ms 8.721 ms
3 142.250.210.33 (142.250.210.33) 15.231 ms 14.982 ms 15.004 ms
4 142.250.183.14 (142.250.183.14) 20.341 ms 20.112 ms 20.276 ms
mshiv@localhost:~$

```
mshiv@localhost:~$ hostname
localhost
mshiv@localhost:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    100    0        0 wlp2s0
192.168.1.0     0.0.0.0         255.255.255.0   U     100    0        0 wlp2s0
mshiv@localhost:~$ netstat -n
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 192.168.1.8:52134       142.250.183.14:443      ESTABLISHED
tcp        0      0 192.168.1.8:52140       142.250.183.78:443      ESTABLISHED
udp        0      0 0.0.0.0:68              0.0.0.0:*
mshiv@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
2: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 10
    link/ether 3c:52:82:9a:4f:1d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic wlp2s0
       valid_lft 86392sec preferred_lft 86392sec
    inet6 fe80::97e4:98d2:62c1:1b2e/64 scope link
       valid_lft forever preferred_lft forever
mshiv@localhost:~$ nslookup google.com
Server:             192.168.1.1
Address:            192.168.1.1#53

Non-authoritative answer:
Name:  google.com
Address: 142.250.183.14
mshiv@localhost:~$ ifconfig
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.8  netmask 255.255.255.0  broadcast 192.168.1.255
        ether 3c:52:82:9a:4f:1d  txqueuelen 1000  (Ethernet)
        RX packets 45231  bytes 62347421 (59.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 29811  bytes 4021536 (3.8 MiB)
        TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
mshiv@localhost:~$ ping -c 3 google.com
PING google.com (142.250.183.14) 56(84) bytes of data.
64 bytes from 142.250.183.14: icmp_seq=1 ttl=116 time=21.3 ms
64 bytes from 142.250.183.14: icmp_seq=2 ttl=116 time=20.8 ms
64 bytes from 142.250.183.14: icmp_seq=3 ttl=116 time=21.0 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 20.812/21.054/21.324/0.208 ms
```


mshiv@localhost: ~
mshiv@localhost:~$ ip
Usage: ip [OPTIONS] OBJECT { COMMAND | help }
ip [-force | -batch filename
where OBJECT := { link | address | addrlabel | route | rule | neigh | ntable |
tunnel | tuntap | maddress | mroute | mrule | monitor | xfrm |
netns | l2tp | fou | macsec | tcp_metrics | token | netconf | ila |
vrf | sr | nexthop }
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
-h[uman-readable] | -iec | -j[son] | -p[retty] |
-f[amily] { inet | inet6 | mpls | bridge | link } |
-4 | -6 | -I | -D | -M | -B | -0
-l[oops] { maximum-addr-flush-attempts } | -br[ief] |
-o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
-rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
-c[olor] }
mshiv@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.00.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
link/ether 00:0c:29:75:99:27 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic ens33
valid_lft 86392sec preferred_lft 86392sec
inet6 fe80::20c:29ff:fe75:9927/64 scope link
valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:9a:0c:8a:ee brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
valid_lft forever preferred_lft forever
mshiv@localhost:~$ ip add show dev dns33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
link/ether 00:0c:29:75:99:27 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.8/24 brd 192.168.1.255 scope global dynamic ens33
valid_lft 1740sec preferred_lft 1740sec
inet6 fe80::20c:29ff:fe75:9927/64 scope link
mshiv@localhost:~$